Team Caffeinated Nerds · Challenge: **Frontier · PS-04** · Fast and Secure Online Transaction Authentication

Bhavyam Misra
Roll No: 1024030535

Gunagye Jain
Roll No: 1024030019

Abhijay Nagal
Roll No: 1024170013

Shefali Sharma
Roll No: 1024030284

Problem, Stakeholder & SDGs

> Fraud has moved past OTPs: victims now approve the payment themselves

THE PROBLEM: ASHA, 72, PAYS A SCAMMER WITH HER OWN PIN

Fake "CBI" officer calls → **Opens UPI on her own phone** → **Types her own PIN** → **Approved: ₹85,000 gone**

A PIN checks **who** you are. Nobody checks **what's happening**.

WHO WE SERVE

- **Banks, UPI apps & gateways**
- **Seniors & first-time payers**
- **Decide inside UPI's 15-s window**

EVIDENCE

₹22,495 cr
lost to cyber fraud in India in 2025 (MHA)

~85%
of losses: scams where the victim pays (MHA)

98.5%
of fraud value is in payments above ₹10k (RBI)

PRIMARY · SDG 16
16.4: cut illicit financial flows & organised crime

SUPPORTING · SDG 8
8.10: access to financial services for all

SUPPORTING · SDG 9
9.1: reliable, resilient (digital) infrastructure

Sources: MHA/NCRP data via Parliament replies (2024, 2025); RBI Discussion Paper on safeguards in digital payments, 9 Apr 2026. Links: repo › docs/02-research/problem-evidence.md

Existing Approaches & Gap

> Today's defences check the credential or the receiver, never the moment

	Static 2FA *UPI PIN · OTP · 3-D Secure*	Risk engines *Stripe Radar · Sift · BioCatch*	India's tools *NPCI FRM · DoT FRI · MuleHunter*	Pramaan *this project*
Checks who holds the credential	✓	✓	✗	✓
Scores the receiver's reputation	✗	✓	✓	✓
Reads the live moment: call, device, typing	✗	—	✗	✓
Explains the decision to the user	✗	✗	✗	✓
Picks the right challenge for the risk	✗	✗	✗	✓
Affordable for small banks & apps	✓	✗	✓	✓

✓ yes — partly ✗ no

The gap: nothing lightweight reads the moment and answers with an explained, right-sized challenge.

Sources: RBI (25 Sep 2025, 9 Apr 2026); Stripe Radar, Sift, BioCatch docs; NPCI FRM; DoT FRI; MuleHunter.AI. Links: repo › docs/02-research/prior-art.md

Proposed Solution

> Pramaan: a PIN checks who you are. We check what's happening, before every payment.

WHAT WE BUILD: A RISK CHECK BEFORE THE PIN

Bank / UPI app calls us → **Scores 20+ live signals** → **Decision in ~40 ms**

WHY IT'S DIFFERENT

- Judged against each user's own habits
- Amount-aware: a ₹40 tea is never stopped
- A plain-language reason with every decision

An OTP can't stop a coached victim. A pause and a family member who can cancel (**n8n → Telegram**) can, **used only when the context is risky.**

THE SAME ₹85,000 PAYMENT · FOUR SITUATIONS

Usual payee, own phone, normal typing	risk 0.04	ALLOW
New phone, fresh SIM, new city	risk 0.58	BIOMETRIC
On a call, new UPI ID pasted	risk 0.71	HOLD + SON
Emulator + VPN, 15 tries in 10 min	risk 0.97	BLOCK

Why this decision

analyst feedback retrains

On a phone call	+0.31
New UPI ID pasted	+0.24
6× usual amount	+0.13
Own device, home	−0.07

Technical Approach

> One API call: context in, an explained decision out, in milliseconds

1 · INPUT

Payment app + SDK

amount · device · typing · paste · on-call*

2 · API

/v1/authenticate

FastAPI · over 150 ms → falls back to 2FA

3 · FEATURES

20+ features

vs the user's own baseline: “8× usual amount”

4 · RISK ENGINE

- a Rules: hard policy + fast-path allow
- b LightGBM → fraud probability p
- c TreeSHAP → each signal's weight
- d Risk type: impostor · scam · bot
- e Cost policy → decision + challenge

5 · OUTPUT

- ALLOW
- CHALLENGE
- BLOCK

+ score · reasons · latency

POWERED BY n8n *guardian loop*

Scam-risk HOLD → n8n workflow triggered → Telegram: Approve / Cancel → No reply in 30 min → cancel → Outcome → training label

STACK · ₹0 RUNNING COST

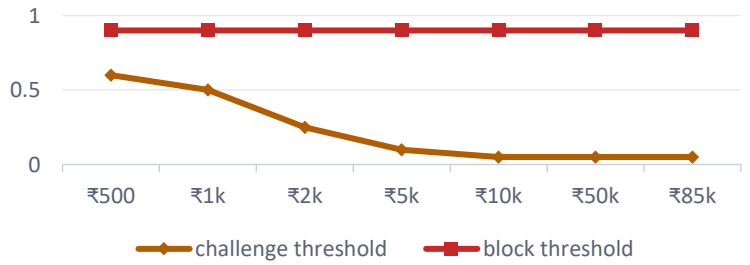
- Python · FastAPI
- LightGBM + SHAP
- React dashboard
- Supabase
- n8n + Telegram
- WebAuthn passkeys

SYNTHETIC DATA (NO PUBLIC SET HAS THESE SIGNALS)

5k	30	4
users	days	attack types

+ hard negatives (travel, new phone) · checked on IEEE-CIS & PaySim

Decision rule: stricter as the amount grows



Build Plan, Risks & Impact

> A deployed, working prototype in the hack, honest about its limits

1 · WHAT WE BUILD

- Live API with fail-safe to 2FA
- Rules + LightGBM + TreeSHAP
- Passkey fingerprint check
- n8n + Telegram trusted-contact loop
- Dashboard with “why” panel

24-HOUR PLAN

Data	Engine	App + UI	Deploy
H0–6	H6–12	H12–18	H18–24

Cost: ₹0, free tiers + open source

2 · RISKS & HOW WE HANDLE THEM

Synthetic data

→ hard negatives, per-type metrics, public-set checks

Call state needs an SDK

→ missing = “unknown”, never “safe”

False positives

→ block only at p ≥ 0.9; ≤ 5% challenged

Privacy (DPDP)

→ derived features only, hashed IDs

3 · EXPECTED IMPACT

Honest payments stay instant.
Risky ones get stopped.

≥ 80%
scams caught, ≤ 5% of honest payments challenged

≥ 90%
honest ₹10k+ transfers with no delay

< 50 ms
p95 decision time (UPI allows 15 s)

Fast. Secure. Explained.